

RAVADA KARAN

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[Linkedin](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering; CGPA: 7.18/10

Chennai, India

2022 – 2026

Narayana Junior College

Class XII (Higher Secondary Certificate - MPC); Percentage: 75.7%

Andhra Pradesh, India

2020 – 2022

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, PHP, SQL, HTML5, CSS3

Frameworks & Libraries: FastAPI, Flask, Streamlit, OpenCV, YOLOv8, Scikit-Learn, Pandas, NumPy

Databases & Developer Tools: MySQL, PostgreSQL, SQLite, Git, GitHub, RESTful APIs, Postman

Core Competencies: Machine Learning Pipelines, Computer Vision, Full-Stack Web Development, OOP

EXPERIENCE

Navodita Infotech

Full-Stack Web Developer

Remote, India

Dec 2025 – Jan 2026

- Engineered and maintained responsive, full-stack web applications utilizing modern front-end and back-end frameworks.
- Designed and integrated secure RESTful APIs to facilitate seamless data communication with relational database systems.
- Optimized database queries and front-end rendering, significantly reducing API latency and improving page load responsiveness.
- Collaborated in Agile sprints, taking part in code reviews, integration testing, and bug tracking to maintain high software reliability.

PROJECTS

Intelligent Traffic Management System | Python, YOLOv8, FastAPI, OpenCV, NumPy

- Constructed a real-time edge AI traffic control system using YOLOv8 for vehicle classification and density tracking.
- Engineered dynamic signal timing algorithms based on real-time vehicle density to mitigate urban intersection congestion.
- Implemented an automated ambulance priority override protocol that clears transit corridors with zero human intervention.
- Developed asynchronous REST APIs using FastAPI for live video stream telemetry processing and traffic monitoring.

Multiple Disease Prediction System | Python, Scikit-Learn, Streamlit, Pandas

- Trained and deployed diagnostic ML models (Random Forest, SVM, Logistic Regression) predicting Diabetes, Heart Disease, and Parkinson's.
- Built an intuitive Streamlit web interface integrated with serialized models for sub-second risk factor evaluations.
- Implemented rigorous data preprocessing, feature engineering, and cross-validation pipelines to improve model sensitivity.

Inventory Management System | Python, Flask, SQLite, HTML, CSS

- Developed a web-based inventory control application with complete CRUD functionality and role-based access control.
- Automated stock tracking, reorder alert notifications, and historical transaction logs to streamline inventory audits.

PUBLICATIONS

Real-Time Intelligent Traffic Signal Control Using YOLOv8 with Ambulance Priority | SSRN Electronic Journal

- Published research proposing an AI-driven adaptive traffic signal framework integrating computer vision (YOLOv8) and dynamic queuing theory for prioritized emergency vehicle clearance.

CERTIFICATIONS & ACHIEVEMENTS

Anthropic Education: AI Fluency: Framework & Foundations (Sep 2026)

AWS Academy: Machine Learning Foundations (Feb 2024)

Oracle: Global Certification (Mar 2025)

Hackathons: Participant / Finalist at SRM Hackathon 8.0 (2024) and Layer 0 Hackathon (2023)